Decision Tree: ID3 Algorithm

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Attribute : Outlook:

Values(Outlook) = Sunny, Overcast, Rain

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$S_{sunny} = [2+, 3-] \quad Entropy(S_{sunny}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.971$$

$$S_{overcast} = [4+, 0-] \qquad Entropy(S_{overcast}) = -\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4} = 0$$

$$S_{rainy} = [3+, 2-] \qquad Entropy(S_{rainy}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.971$$

$$Gain(S, outlook) = Entropy(S) - \sum_{v\epsilon(sunny,overcast,rainy)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, outlook) = Entropy(S) - \frac{5}{14}Entropy(S_{sunny}) - \frac{4}{14}Entropy(S_{overcast}) - \frac{5}{14}Entropy(S_{rain})$$

$$Gain(S, outlook) = 0.94 - \frac{5}{4}0.971 - \frac{4}{14}0 - \frac{5}{14}0.971 = 0.2464$$

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Attribute: Temp

Values(Temp)=Hot, Mild, Cool

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$S_{Hot} = [2+, 2-] \quad Entropy(S_{Hot}) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 1.0$$

$$S_{Mild} = [4+, 2-] \qquad Entropy(S_{Mild}) = -\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6} = 0.9183$$

$$S_{Cool} = [3+, 1-] \qquad Entropy(S_{Cool}) = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} = 0.8113$$

$$Gain(S, temp) = Entropy(S) - \sum_{v\epsilon(Hot,Mild,Cool)} \frac{|S_v|}{|S|} Entropy(S_v)\Big)$$

$$Gain(S, temp) = Entropy(S) - \frac{4}{14}Entropy(S_{Hot}) - \frac{6}{14}Entropy(S_{Mild}) - \frac{4}{14}Entropy(S_{Cool}\Big)$$

$$Gain(S, temp) = 0.94 - \frac{4}{14}1.0 - \frac{6}{14}0.9183 - \frac{4}{14}0.8113 = 0.0289$$

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Attribute: Humidity

Values(Humidity)=High, Normal

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$S_{High} = [3+, 4-] \quad Entropy(S_{High}) = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7} = 0.9852$$

$$S_{Normal} = [6+, 1-] \quad Entropy(S_{Normal}) = -\frac{6}{7}\log_2\frac{6}{7} - \frac{1}{7}\log_2\frac{1}{7} = 0.5916$$

$$Gain(S, Humidity) = Entropy(S) - \sum_{v\epsilon(High,Normal)} \frac{|S_v|}{|S|} Entropy(S_v)\Big)$$

$$Gain(S, Humidity) = Entropy(S) - \frac{7}{14}Entropy(S_{High}) - \frac{7}{14}Entropy(S_{Normal})$$

$$Gain(S, Humidity) = 0.94 - \frac{7}{14}0.9852 - \frac{7}{14}0.5916 - \frac{4}{14}0.8113 = 0.1576$$

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Attribute: Wind

Values(Wind)=Strong, Weak

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$S_{Strong} = [3+, 3-] \quad Entropy(S_{Strong}) = 1.0$$

$$S_{Weak} = [6+, 2-] \qquad Entropy(S_{Weak}) = -\frac{6}{8}\log_2\frac{6}{8} - \frac{2}{8}\log_2\frac{2}{8} = 0.8113$$

$$Gain(S, Wind) = Entropy(S) - \sum_{v\epsilon(Strong,Weak)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, Wind) = Entropy(S) - \frac{6}{14} Entropy(S_{Strong}) - \frac{8}{14} Entropy(S_{Weak})$$

$$Gain(S, Wind) = 0.94 - \frac{6}{14}1.0 - \frac{8}{14}0.8113 - \frac{4}{14}0.8113 = 0.0478$$

$$Gain(S, Outlook) = 0.2364$$

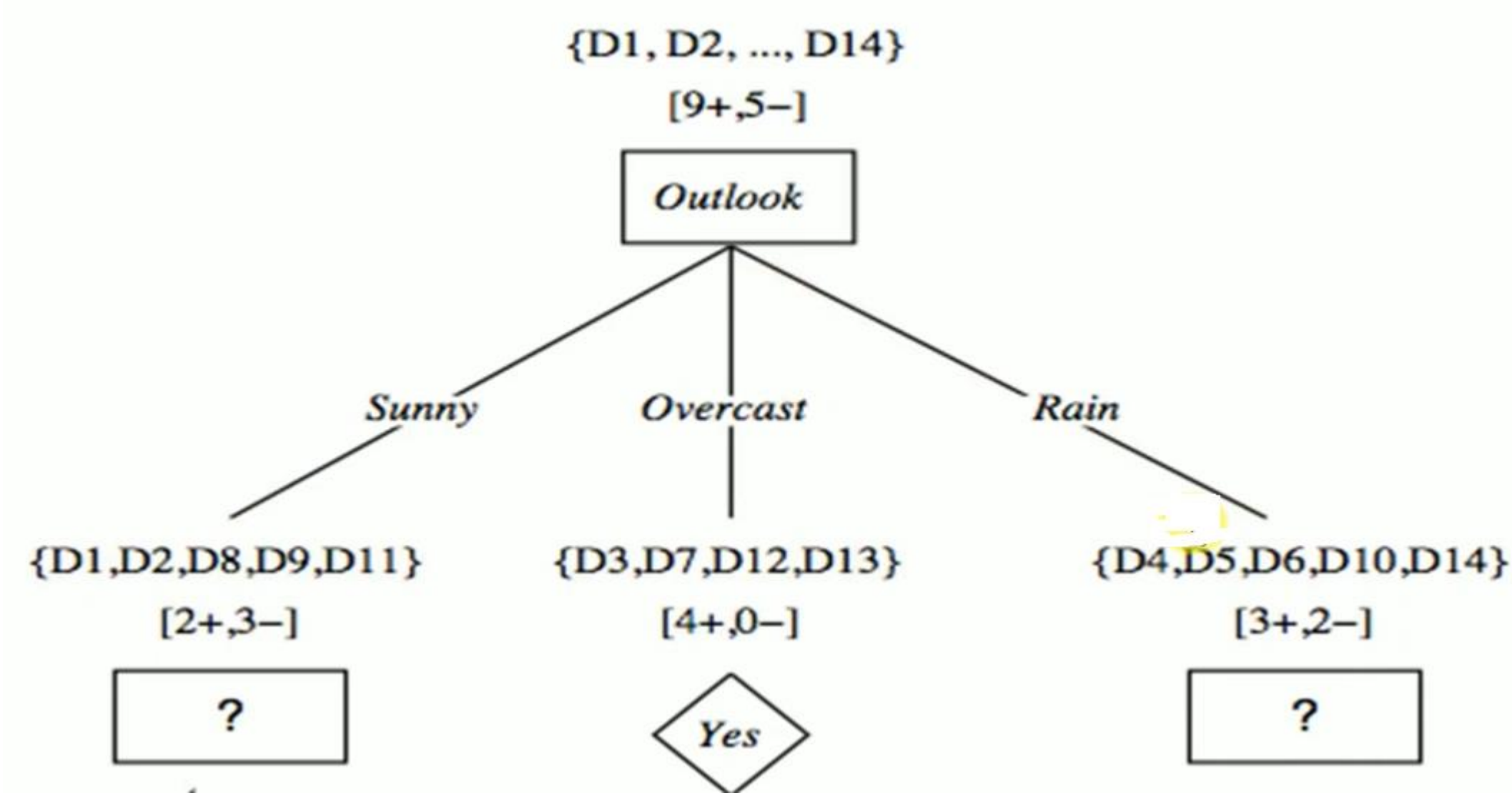
$$Gain(S, Temp) = 0.0289$$

$$Gain(S, Humidity) = 0.1576$$

$$Gain(S, Wind) = 0.0479$$

Maximum Information Gain is of Outlook, that is =0.2464

Now the tree will be as follows:



Now Considering Sunny:

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Temp

Values(Temp)=Hot, Mild, Cool

$$S_{sunny} = [2+, 3-] \quad Entropy(S) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$$

$$S_{Hot} = [0+, 2-] \quad Entropy(S_{Hot}) = 0.0$$

$$S_{Mild} = [1+, 1-] \qquad Entropy(S_{Mild}) = 1.0$$

$$S_{Cool} = [1+, 0-] \qquad Entropy(S_{Cool}) = 0.0$$

$$Gain(S_{sunny}, temp) = Entropy(S) - \sum_{v\epsilon(Hot,Mild,Cool)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{sunny}, temp) = Entropy(S) - \frac{2}{5}Entropy(S_{Hot}) - \frac{2}{5}Entropy(S_{Mild}) - \frac{1}{5}Entropy(S_{Cool})$$

$$Gain(S_{sunny}, temp) = 0.97 - \frac{2}{5}0.0 - \frac{2}{5}1 - \frac{1}{5}0.0 = 0.570$$

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Humidity

Values(Humidity)=High, Normal

$$S_{sunny} = [2+, 3-] \quad Entropy(S) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$$

$$S_{High} = [0+, 3-] \quad Entropy(S_{High}) = 0.0$$

$$S_{Normal} = [2+, 0-] \qquad Entropy(S_{Normal}) = 0.0$$

$$Gain(S_{sunny}, Humidity) = Entropy(S) - \sum_{v\epsilon(High,Normal)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{sunny}, Humidity) = Entropy(S) - \frac{3}{5}Entropy(S_{High}) - \frac{2}{5}Entropy(S_{Normal})$$

$$Gain(S_{sunny}, Humidity) = 0.97 - \frac{3}{5}0.0 - \frac{2}{5}0.0 = 0.97$$

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Wind

Values(Wind)=Strong, Weak

$$S_{sunny} = [2+, 3-] \quad Entropy(S) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$$

$$S_{Strong} = [1+, 1-] \quad Entropy(S_{Strong}) = 1.0$$

$$S_{Weak} = [1+, 2-] \qquad Entropy(S_{Weak}) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} = 0.9183$$

$$Gain(S_{sunny}, Wind) = Entropy(S) - \sum_{v\epsilon(Strong,Weak)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{sunny}, Wind) = Entropy(S) - \frac{2}{5} Entropy(S_{Strong}) - \frac{3}{5} Entropy(S_{Weak})$$

$$Gain(S, Wind) = 0.94 - \frac{2}{5}1.0 - \frac{3}{5}0.9183 = 0.0192$$

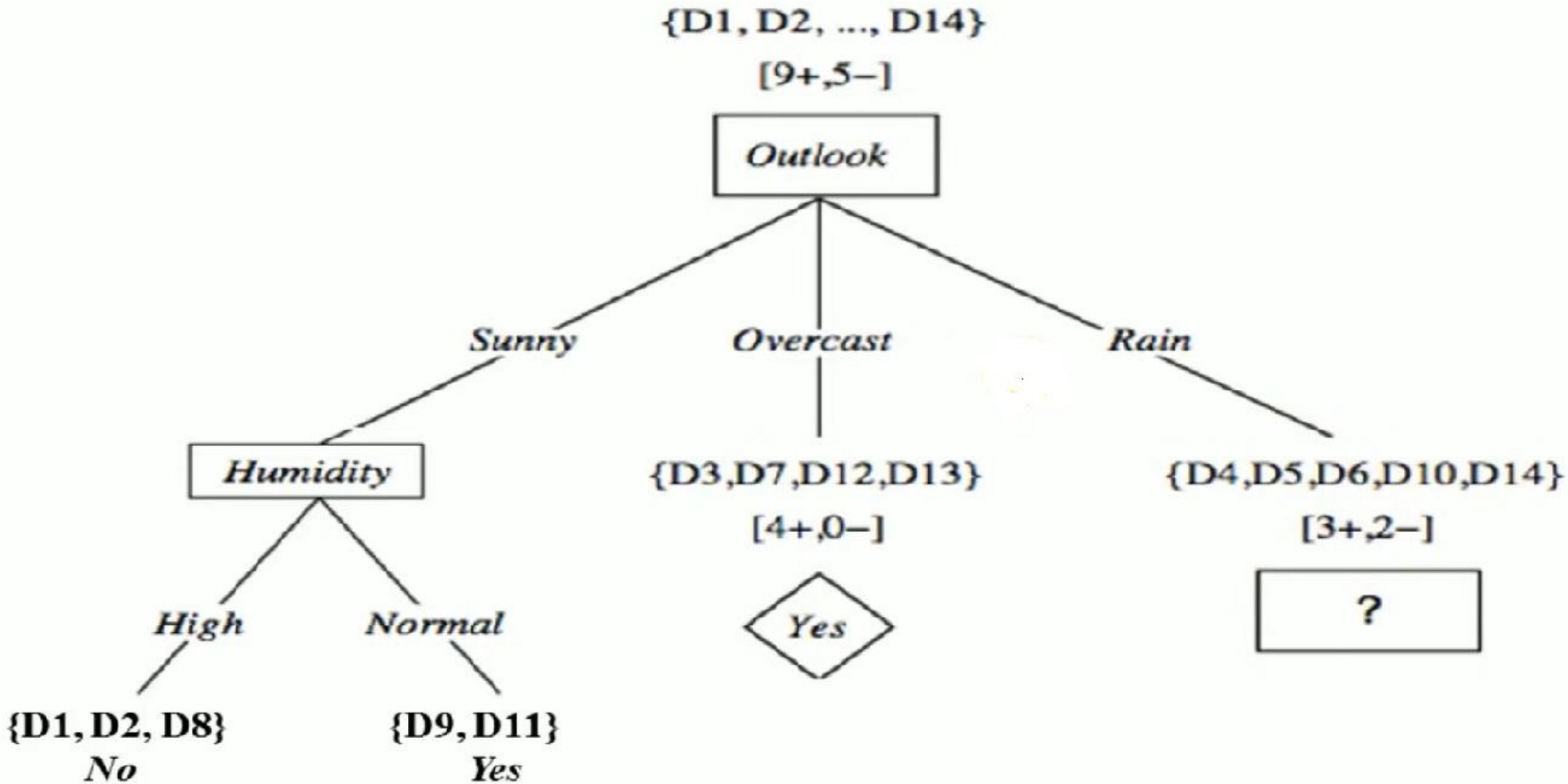
$$Gain(S_{sunny}, Temp) = 0.570$$

$$Gain(S_{sunny}, Humidity) = 0.97$$

$$Gain(S_{sunny}, Wind) = 0.0192$$

Maximum Information Gain is of Humidity, that is 0.97

So the tree can be drawn as:



Now For Rain :

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Temp

Values(Temp)=Hot, Mild, Cool

$S_{Rainy} = [3+, 2-] \quad Entropy(S_{Rainy}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$

$S_{Hot} = [0+, 0-] \quad Entropy(S_{Hot}) = 0.0$

$S_{Mild} = [2+, 1-] \qquad Entropy(S_{Mild}) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.9183$

$S_{Cool} = [1+, 1-] \qquad Entropy(S_{Cool}) = 1.0$

$$Gain(S_{rainy}, temp) = Entropy(S) - \sum_{v\epsilon(Hot,Mild,Cool)} \frac{|S_v|}{|S|} Entropy(S_v)\Big)$$

$$Gain(S_{rainy}, temp) = Entropy(S) - \frac{0}{5}Entropy(S_{Hot}) - \frac{3}{5}Entropy(S_{Mild}) - \frac{2}{5}Entropy(S_{Cool}\Big)$$

$$Gain(S_{rainy}, temp) = 0.97 - \frac{0}{5}0.0 - \frac{3}{5}0.98 - \frac{2}{5}1.0 = 0.0192$$

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Humidity

Values(Humidity)=High, Normal

$$S_{rainy} = [3+, 2-] \quad Entropy(S_{rainy}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$$

$$S_{High} = [1+, 1-] \quad Entropy(S_{High}) = 1.0$$

$$S_{Normal} = [2+, 1-] \qquad Entropy(S_{Normal}) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.9183$$

$$Gain(Rainy, Humidity) = Entropy(S) - \sum_{v\epsilon(High,Normal)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(Rainy, Humidity) = Entropy(S) - \frac{2}{5}Entropy(S_{High}) - \frac{3}{5}Entropy(S_{Normal})$$

$$Gain(Rainy, Humidity) = 0.97 - \frac{2}{5}1.0 - \frac{3}{5}0.918 = 0.0192$$

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Wind

Values(Wind)=Strong, Weak

$$S_{rainy} = [3+, 2-] \quad Entropy(S_{rainy}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$$

$$S_{Strong} = [0+, 2-] \quad Entropy(S_{Strong}) = 0.0$$

$$S_{Weak} = [3+, 0-] \qquad Entropy(S_{Weak}) = 0.0$$

$$Gain(S_{rainy}, Wind) = Entropy(S) - \sum_{v\epsilon(Strong,Weak)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{rainy}, Wind) = Entropy(S) - \frac{2}{5}Entropy(S_{Strong}) - \frac{3}{5}Entropy(S_{Weak})$$

$$Gain(S_{rainy}, Wind) = 0.97 - \frac{2}{5}0.0 - \frac{3}{5}0.0 = 0.97$$

$$Gain(S_{rainy}, Temp) = 0.0192$$

$$Gain(S_{rainy}, Humidity) = 0.0192$$

$$Gain(S_{rainy}, Wind) = 0.97$$

Therefore 0.97 is maximum

Wind will be considered at that level.

So the final tree will be:

